

INFO90002 S1 2026 Assignment 2 - SQL

Hospital Management System

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Table suffix used: 6981

Note: SQL statements use the ENG script table names with suffix 6981.

Q1. List all types of rooms each department has. Your list should show department ID and name and room type ordered alphabetically by department name.

SQL statement:

```
SELECT DISTINCT
  '1856981' AS StuID,
  d.dept_Id AS DepartmentID,
  d.dept_Name AS DepartmentName,
  r.room_Type AS RoomType
FROM Department6981 AS d
INNER JOIN Room6981 AS r
  ON d.dept_Id = r.dept_Id
ORDER BY d.dept_Name, r.room_Type;
```

Result screenshot:

	StuID	DepartmentID	DepartmentName	RoomType
	1856981	120	Anesthesiology	Consultation Room
	1856981	120	Anesthesiology	Deluxe Room
	1856981	120	Anesthesiology	Emergency Room
	1856981	120	Anesthesiology	Standard Room
	1856981	120	Anesthesiology	Super Deluxe Room
	1856981	126	Audiology & Speech	Consultation Room
	1856981	126	Audiology & Speech	Deluxe Room
	1856981	126	Audiology & Speech	Standard Room
	1856981	126	Audiology & Speech	Super Deluxe Room
	1856981	101	Cardiology	Consultation Room

145 rows returned

Q2. List all departments, wards they contain and number of beds in each ward that have at least 10 beds per ward. Your list should display department ID, department name, ward number, ward name and total number of beds per ward in the descending order of number of beds.

SQL statement:

```
SELECT
    '1856981' AS StuID,
    d.dept_ID,
    d.dept_Name,
    w.ward_No,
    w.ward_Name,
    COUNT(b.bed_No) AS Total_Bed
FROM Department6981 AS d
INNER JOIN Ward6981 AS w
    ON d.dept_ID = w.dept_ID
INNER JOIN Bed6981 AS b
    ON w.ward_No = b.ward_No
GROUP BY d.dept_ID,
         d.dept_Name,
         w.ward_No,
         w.ward_Name
HAVING COUNT(bed_No) >= 10
ORDER BY Total_Bed DESC;
```

Result screenshot:

StuID	dept_ID	dept_Name	ward_No	ward_Name	Total_Bed
1856981	117	Surgery	538	Surgical Ward	16
1856981	101	Cardiology	501	ICU	15
1856981	103	Gastroenterology	510	General Medicine Ward	14
1856981	102	Neurology	508	ICU	12
1856981	118	Radiology	540	General Medicine Ward	12
1856981	104	Nephrology	512	General Medicine Ward	12
1856981	105	Pulmonology	513	General Medicine Ward	10
1856981	107	General Medicine	519	General Medicine Ward	10
1856981	116	ENT	536	Surgical Ward	10
1856981	117	Surgery	537	Surgical Ward	10

19 rows returned

Q3. List doctors who had appointments in January of the last year. Results should show doctor full name, appointment date, patient's last name and patient ID, ordered by doctor ID and appointment date.

SQL statement:

```
SELECT
    '1856981' AS StuID,
    CONCAT(d.FName, ' ', d.LName) AS DoctorName,
    a.appointment_Date AS AppointmentDate,
    p.LName AS Patient_LastName,
    p.patient_Id AS PatientID
FROM Appointment6981 AS a
INNER JOIN Patients6981 AS p
    ON a.patient_Id = p.patient_Id
INNER JOIN Doctor6981 AS d
    ON a.doct_Id = d.doct_Id
WHERE YEAR(CURDATE()) - 1 = YEAR(a.appointment_Date) AND MONTH(a.appointment_Date) = 1
GROUP BY d.doct_Id,
         CONCAT(d.FName, ' ', d.LName),
         p.patient_Id,
         p.LName,
         a.appointment_Date
ORDER BY d.doct_Id, a.appointment_Date;
```

Result screenshot:

StuID	DoctorName	AppointmentD...	Patient_LastName	PatientID
1856981	Dr. Najeab Basir	2025-01-11	Naqvi	658
1856981	Dr. Najeab Basir	2025-01-20	Mughal	512
1856981	Dr. Najeab Basir	2025-01-26	Syed	641
1856981	Dr. Kamran Ahmed	2025-01-13	Saeed	1573
1856981	Dr. Hanif Kamal	2025-01-11	Hakim	251
1856981	Dr. Hanif Kamal	2025-01-22	Fayyaz	1226
1856981	Dr. Hanif Kamal	2025-01-29	Mughal	366
1856981	Dr. Tahir Hamid	2025-01-01	Mehmood	566
1856981	Dr. Tahir Hamid	2025-01-08	Saeed	1059
1856981	Dr. Tahir Hamid	2025-01-16	Ali	240

76 rows returned

Q4. List doctors with most expensive appointments. Results should show doctor's name as last name, comma, first name, appointment date, patient name as first name then last name, and payment amount.

SQL statement:

```
SELECT
    '1856981' AS StuID,
    CONCAT(d.LName,',',d.FName) AS Doctor_Name,
    a.appointment_Date AS Appointment_Date,
    CONCAT(p.FName,' ',p.LName) AS Patient_Name,
    a.payment_amount AS The_Most_Expensive_Amount
FROM Appointment6981 AS a
INNER JOIN Patients6981 as p
    ON a.patient_Id = p.patient_Id
INNER JOIN Doctor6981 AS d
    ON d.doct_Id = a.doct_Id
WHERE
    a.payment_amount = (
        SELECT MAX(a.payment_amount)
        FROM Appointment6981 AS a
    )
ORDER BY a.payment_amount DESC;
```

Result screenshot:

StuID	Doctor_Name	Appointment_Date	Patient_Name	The_Most_Expensive_Amount
1856981	Hasnain,Dr. Raza	2025-02-16	Khalid Ashraf	2499
1856981	Hasnain,Dr. Raza	2025-06-13	Hussain Mazhar	2499

2 rows returned

Q5. List all wards that currently have no beds. Results should show ward number and ward name sorted in alphabetical order of ward names.

SQL statement:

```
SELECT
    '1856981' AS StuID,
    w.ward_No AS Ward_No,
    w.ward_Name AS Ward_Name
FROM Ward6981 AS w
LEFT JOIN Bed6981 AS b
    ON w.ward_No = b.ward_No
GROUP BY
    w.ward_No,
    w.ward_Name
HAVING COUNT(b.bed_No) = 0
ORDER BY w.ward_Name;
```

Result screenshot:

StuID	Ward_No	Ward_Name
1856981	563	Geriatrics Ward

1 row returned

Q6. List how many rooms have never been used within each department, not including consultation rooms. For each department with unused rooms, show department ID, department name and number of unused rooms in descending order of room count.

SQL statement:

```
SELECT
    '1856981' AS StuID,
    d.dept_Id AS Department_Id,
    d.dept_Name AS Department_Name,
    COUNT(r.room_No) AS Unused_Rooms
FROM Department6981 AS d
INNER JOIN Room6981 AS r
    ON r.dept_Id = d.dept_Id
LEFT JOIN RoomRecords6981 AS rr
    ON r.room_No = rr.room_no
WHERE r.room_Type <> 'Consultation Room'
    AND rr.room_no IS NULL
GROUP BY
    d.dept_Id,
    d.dept_Name
HAVING COUNT(r.room_No) > 0
ORDER BY Unused_Rooms DESC;
```

Result screenshot:

	StuID	Department_Id	Department_Name	Unused_Rooms
	1856981	117	Surgery	30
	1856981	130	Critical Care	3
	1856981	107	General Medicine	3
	1856981	108	Pediatrics	3
	1856981	121	Emergency Medicine	3
	1856981	129	Family Medicine	2
	1856981	116	ENT	2
	1856981	114	Psychology	1
	1856981	127	Homeopathy	1
	1856981	125	Immunology	1

25 rows returned

Q7. What is the average length of a surgical procedure, for surgeries where the patient had already presented with diabetes? Show the time in minutes.

SQL statement:

```
SELECT
    '1856981' AS StuID,
    AVG(TIMESTAMPDIFF(
        MINUTE,
        TIMESTAMP(s.surgery_Date, s.start_Time),
        TIMESTAMP(s.surgery_Date, s.end_Time))) AS Average_Surgery_Length_Minutes
FROM Patients6981 AS p
INNER JOIN SurgeryRecord6981 AS s
    ON s.patient_Id = p.patient_Id
INNER JOIN MedicalRecord6981 AS m
    ON m.patient_Id = p.patient_Id
WHERE LOWER(m.diagnosis) LIKE '%diabetes%'
    AND s.surgery_Date > m.visit_Date;
```

Result screenshot:

	StuID	Average_Surgery_Length_Minu...
	1856981	30.0000

1 row returned

Q8. List surgeries where any two of doctor, nurse or helper had the same last name. Show surgeon ID and the doctor's, nurse's and helper's full names; order by doctor's name.

SQL statement:

```
SELECT
    '1856981' AS StuID,
    d.doct_Id AS DoctorID,
    CONCAT(d.FName, ' ', d.LName) AS DoctorName,
    CONCAT(n.FName, ' ', n.LName) AS NurseName,
    CONCAT(h.FName, ' ', h.LName) AS HelperName
FROM SurgeryRecord6981 AS s
INNER JOIN Helpers6981 AS h
    ON h.helper_Id = s.helper_Id
INNER JOIN Nurse6981 AS n
    ON n.nurse_Id = s.nurse_Id
INNER JOIN Doctor6981 AS d
    ON d.doct_Id = s.surgeon_ID
WHERE d.LName = n.LName
    OR d.LName = h.LName
    OR n.LName = h.LName
ORDER BY CONCAT(d.FName, ' ', d.LName);
```

Result screenshot:

	StuID	DoctorID	DoctorName	NurseName	HelperName
	1856981	1153	Dr. Aamir Jalil	Sohail Mian	Iftikhar Mian
	1856981	1150	Dr. Abu Bakar Hafeez Bhatti	Ijaz Usmani	Sameer Usmani
	1856981	1150	Dr. Abu Bakar Hafeez Bhatti	Irum Sheikh	Inam Sheikh
	1856981	1159	Dr. Ajmal Hasan Naqvi	Lateef Usmani	Liaqat Usmani
	1856981	1164	Dr. Akbar Khan	Inam Butt	Nadeem Butt
	1856981	1164	Dr. Akbar Khan	Rameez Khan	Liaqat Dar
	1856981	1164	Dr. Akbar Khan	Yusra Malik	Rabia Malik
	1856981	1180	Dr. Ayesha Mahrukh	Nazia Butt	Laila Butt
	1856981	1166	Dr. Bilal Asghar	Bushra Usmani	Hassan Usmani
	1856981	1165	Dr. Javaid Iqbal	Rubina Lodhi	Shaista Lodhi

27 rows returned

Q9. List numbers of doctors, nurses and helpers in each department and the total number of medical staff. Show department name, medical staff type, and number of staff; last row should show empty department name, 'Total', and total value.

SQL statement:

```
SELECT
    '1856981' AS StuID,
    dp.dept_Name AS DepartmentName,
    'Doctor' AS MedStaff,
    COUNT(dt.doct_Id) AS Number
FROM Department6981 AS dp
INNER JOIN Doctor6981 AS dt
    ON dt.dept_Id = dp.dept_Id
GROUP BY dp.dept_Name
UNION ALL
SELECT
    '1856981' AS StuID,
    dp.dept_Name AS DepartmentName,
    'Nurse' AS MedStaff,
    COUNT(n.nurse_Id) AS Number
FROM Department6981 AS dp
INNER JOIN Nurse6981 AS n
    ON dp.dept_Id = n.dept_Id
GROUP BY dp.dept_Name
UNION ALL
SELECT
    '1856981' AS StuID,
    dp.dept_Name AS DepartmentName,
    'Helper' AS MedStaff,
    COUNT(h.helper_Id) AS Number
FROM Department6981 AS dp
INNER JOIN Helpers6981 AS h
    ON dp.dept_Id = h.dept_Id
GROUP BY dp.dept_Name
UNION ALL
SELECT
    '1856981' AS StuID,
    '' AS DepartmentName,
    'Total' AS MedStaff,
    (SELECT COUNT(*) FROM Doctor6981)
    + (SELECT COUNT(*) FROM Nurse6981)
    + (SELECT COUNT(*) FROM Helpers6981) AS Number;
```

Result screenshot:

	StuID	DepartmentNa...	MedStaff	Number
	1856981	Nutrition & Diete...	Helper	37
	1856981	Rehabilitation	Helper	33
	1856981	Hematology	Helper	43
	1856981	Immunology	Helper	40
	1856981	Audiology & Sp...	Helper	27
	1856981	Homeopathy	Helper	27
	1856981	Sexology	Helper	40
	1856981	Family Medicine	Helper	29
	1856981	Critical Care	Helper	30
	1856981	Geriatrics	Helper	43
	1856981		Total	2000

93 rows returned

Q10a. Add a column capacity to Room6981. Show all columns with the respective data types using the relevant SQL statement.

SQL statement:

```
ALTER TABLE Room6981
ADD COLUMN capacity INT;
```

```
DESCRIBE Room6981;
```

Result screenshot:

	Field	Type	Null	Key	Default	Extra
	room_No	int	NO	PRI	NULL	
	dept_Id	int	YES	MUL	NULL	
	room_Type	varchar(100)	YES		NULL	
	capacity	int	YES		NULL	

4 rows returned Q10b. Set capacity of all standard rooms to 4. Check that all standard rooms got updated by running a SELECT statement with student ID.

SQL statement:

```
UPDATE Room6981
SET capacity = 4
WHERE room_Type = 'Standard Room'
AND room_No > 0;
```

```
SELECT
    '1856981' AS StuID,
    room_No,
    dept_Id,
    room_Type,
    capacity
FROM Room6981
WHERE room_Type = 'Standard Room';
```

Result screenshot:

	StuID	room_No	dept_Id	room_Type	capacity
	1856981	13	101	Standard Room	4
	1856981	16	101	Standard Room	4
	1856981	19	101	Standard Room	4
	1856981	31	102	Standard Room	4
	1856981	34	102	Standard Room	4
	1856981	46	103	Standard Room	4
	1856981	49	103	Standard Room	4
	1856981	61	104	Standard Room	4
	1856981	64	104	Standard Room	4
	1856981	76	105	Standard Room	4

60 rows returned

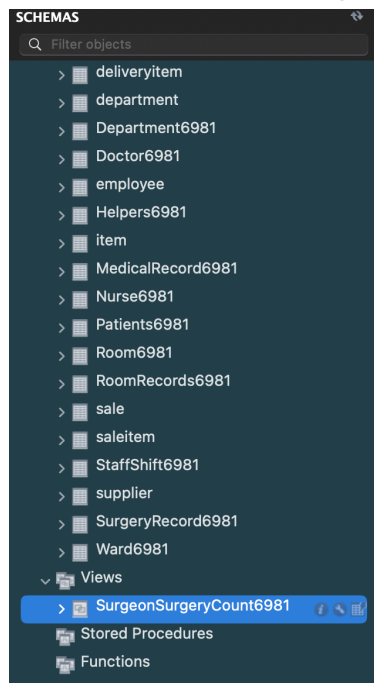
Q11a. Create a view that counts the number of surgeries performed by each surgeon. Include surgeon ID, full name and number of surgeries. Do not include StuID in this view.

SQL statement:

```
CREATE VIEW SurgeonSurgeryCount6981 AS
SELECT
    d.doct_Id AS Surgeon_Id,
    CONCAT(d.FName, ' ', d.LName) AS Surgeon_Name,
    COUNT(s.surgery_Id) AS Number_Of_Surgeries
FROM Doctor6981 AS d
INNER JOIN SurgeryRecord6981 AS s
    ON d.doct_Id = s.surgeon_Id
GROUP BY
    d.doct_Id,
    CONCAT(d.FName, ' ', d.LName);

SELECT *
FROM SurgeonSurgeryCount6981;
```

Workbench left pane showing created view:



SELECT from view result screenshot:

	Surgeon_Id	Surgeon_Name	Number_Of_Surgeri...
	1147	Dr. Nabeel Iqbal	28
	1148	Dr. Rameez Chaudhary	27
	1149	Dr. Imran Hameed	22
	1150	Dr. Abu Bakar Hafeez Bhatti	26
	1151	Dr. Muhammad Faiq	31
	1152	Dr. Talha Abdul Halim	26
	1153	Dr. Aamir Jalil	27
	1154	Dr. Pervez Iqbal	24
	1155	Dr. Zafar Fayyaz	24
	1156	Dr. Muhammad Karamat	31
	1157	Dr. Muhammad Aamir Saghir	22

38 rows returned by SELECT from view

Q11b. Using the view created in Q11a, list the surgeons who completed the highest number of surgeries. Display surgeon ID, full name and number of surgeries, with student ID.

SQL statement:

```
SELECT
    '1856981' AS StuID,
    Surgeon_Id,
    Surgeon_Name,
    Number_Of_Surgeries
FROM SurgeonSurgeryCount6981
WHERE Number_Of_Surgeries = (
    SELECT MAX(Number_Of_Surgeries)
    FROM SurgeonSurgeryCount6981);
```

Result screenshot:

	StuID	Surgeon_Id	Surgeon_Name	Number_Of_Surgeri...
	1856981	1178	Dr. Fauzia Anjum	39

1 row returned